

# Multiply Decimals

Lesson 1-6

Name: \_\_\_\_\_

Date: \_\_\_\_\_

Class: \_\_\_\_\_

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## Key Vocabulary

Level 2 Standard

Picture first, then the word, then a plain-language meaning. Say each word out loud.

$$6 \times 4 = 24$$

Product = answer to multiplication

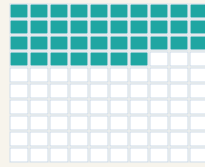
$4 \times 7 = 28$ , so the product is 28

### Product

Write the definition:

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0.37

In 3.25, the decimal point separates 3 (ones) from 25 (tenths and hundredths)

### Decimal point

Write the definition:

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$$38 \times 21$$

$\approx$

$$40 \times 20 = 800$$

about (rounded)

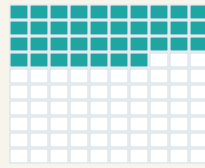
$3.7 \times 4.2$ : round to  $4 \times 4 = 16$ , so the exact answer should be near 16

### Estimate

Write the definition:

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0.37

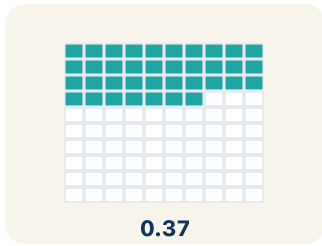
In 0.36: the 3 is in the tenths place (0.3) and the 6 is in the hundredths place (0.06)

### Place value

Write the definition:

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$$3.4 (1 \text{ place}) \times 2.6 (1 \text{ place}) = 8.84 (2 \text{ places total})$$

### Decimal places

Write the definition:

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## Guided Notes

Level 2 Standard



### WHAT WE'RE LEARNING TODAY

**I can multiply decimals and place the decimal point correctly in the product.**



**Fill in each blank as we go. Use the Word Bank to help you.**



### WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Product

Decimal point

Estimate

Place value

Decimal places

**1** The answer to a multiplication problem is the \_\_\_\_\_.

**2** The dot that separates whole numbers from parts of a whole is the \_\_\_\_\_.

**3** Finding an answer that is close but not exact, to check if a product is reasonable, is to \_\_\_\_\_.

**4** The value a digit has based on its position is its \_\_\_\_\_.

**5** To place the point in a product, I count the total number of \_\_\_\_\_ in both factors.



### Watch & Try — Worked Examples

*See the notes in action: watch one worked all the way through, then try the next with the same steps.*

 **I do — watch**

Follow each step as your teacher solves it.

**Problem:** What is  $3.4 \times 2.6$ ?

- A. 8.84
- B. 88.4
- C. 0.884
- D. 8.48

**Step 1**  $34 \times 26 = 884$ .

**Step 2** There are 2 total decimal places ( $1 + 1$ ), so place the decimal to get 8.84.

 **Answer:** A. 8.84

 **We do — together**

Solve this one with your class using the same steps.

**Problem:** What is  $0.7 \times 1.5$ ?

- A. 1.05
- B. 10.5
- C. 0.105
- D. 1.50

**Step 1** \_\_\_\_\_

**Step 2** \_\_\_\_\_

**Answer:** \_\_\_\_\_

 **You do — your turn**

Now try one on your own. Show every step.

**Problem:** What is  $5.3 \times 4$ ?

- A. 21.2
- B. 2.12
- C. 212
- D. 20.12

Show your work:

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## Try It

Solve on your own. Check the answer key when you are done.

**1. Each lemonade uses 0.6 of a cup of sugar, and sugar costs \$0.40 per cup. How much sugar cost is in one lemonade? ( $0.6 \times 0.4$ )**

- A. \$0.24
- B. \$2.40
- C. \$0.024
- D. \$24.00

Show your work:

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**2. The truck sells 2.5 pounds of fries, and each pound earns \$1.20 in profit. What is the total profit? ( $2.5 \times 1.2$ )**

- A. \$3.00
- B. \$30.00
- C. \$0.30
- D. \$3.70

Show your work:

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## Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

**A family buys 3.5 pounds of deli meat at \$6.80 per pound. Show how to find the total cost. Use estimation to check if your answer is reasonable.**

*Sentence starter: The total cost is \$\_\_\_\_ because  $3.5 \times \$6.80 = \underline{\hspace{1cm}}$ . I can estimate:  $\underline{\hspace{1cm}} \times \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$ , which is close to my exact answer, so it is reasonable.*

Show your work:

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## Reflect — Exit Ticket

**What is  $4.6 \times 2.3$ ?**

- A. 10.58
- B. 105.8
- C. 1.058
- D. 10.48

Your answer:

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## Answer Key & Teacher Guide

1. **Notes 1:** Product
2. **Notes 2:** Decimal point
3. **Notes 3:** Estimate
4. **Notes 4:** Place value
5. **Notes 5:** Decimal places
6. **We Try (notes):** A.  $1.05 - 7 \times 15 = 105$ . *There are 2 total decimal places (1 + 1), so place the decimal to get 1.05.*
7. **You Try (notes):** A.  $21.2 - 53 \times 4 = 212$ . *There is 1 total decimal place, so place the decimal to get 21.2.*
8. **Try It 1:** A.  $\$0.24 - 6 \times 4 = 24$ . *There are 2 total decimal places (1 + 1), so  $0.6 \times 0.4 = 0.24$ , which is \$0.24.*
9. **Try It 2:** A.  $\$3.00 - 25 \times 12 = 300$ . *There are 2 total decimal places (1 + 1), so  $2.5 \times 1.2 = 3.00 = \$3.00$ .*
10. **Exit Ticket:** A.  $10.58 - 46 \times 23 = 1,058$ . *There are 2 total decimal places (1 + 1), so the answer is 10.58.*